



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	Quiz 3 SOLUTION	Marks 30 Weight 4%	April 01, 2026
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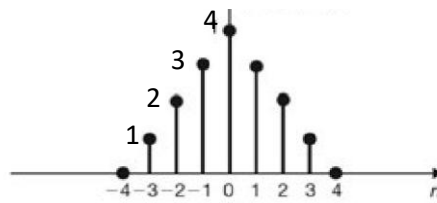
Important Notes

Clearly write your name and student ID on top.

CLO-3	Analyze DT signals and systems in the transform domain using Z transform.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
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Problem 1 [10 pts]

The impulse response of a DT-LTI system is given as



Find its transfer function $H(z)$ using the bilateral Z-Transform and analyze its convergence (i.e., determine the region of convergence).

Solution:

Using definition of the bilateral Z-Tx and noting that $H(z)$ is the Z-Tx of $h[n]$, we get

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n} = z^3 + 2z^2 + 3z + 4 + 3z^{-1} + 2z^{-2} + z^{-3}$$

Which converges for all complex z except $z = 0$ and $z = \infty$.

Problem 2 [10 pts]

A discrete-time LTI system has transfer function

$$H[z] = \frac{z}{z^2 + 7z + 10}$$

Determine the impulse response of the system and also analyze its stability using both time and frequency domain.

Solution:

We apply PFE to $\frac{H[z]}{z}$ leading to

$$\frac{H[z]}{z} = \frac{1}{z^2 + 7z + 10} = \frac{1}{(z+2)(z+5)} = \frac{1}{3} \frac{1}{z+2} - \frac{1}{3} \frac{1}{z+5}$$

Multiplying out by z gives

$$H[z] = \frac{1}{3} \frac{z}{z+2} - \frac{1}{3} \frac{z}{z+5}$$

Using Z-Transform pairs

$$h[n] = \frac{1}{3} (-2)^n u[n] - \frac{1}{3} (-5)^n u[n]$$

Stability test in frequency domain:

System Transfer Function $H(z)$ has two poles, both of which are outside the unit circle, making it unstable.

Stability test in time domain:

The impulse response is not absolutely integrable. Therefore, the system is unstable.

Problem 3 [10 pts]

Suppose $x[k]$ is a causal signal (i.e., $x[k] = 0$ for $k < 0$). Now use

$$\sum_{k=0}^n x[k] = \sum_{k=0}^{\infty} x[k]u[n-k]$$

to show that

$$\sum_{k=0}^n x[k] \iff \frac{zX[z]}{z-1}$$

Solution:

To begin, notice that we can represent the accumulation of $x[n]$ as

$$\sum_{k=0}^n x[k] = \sum_{k=0}^{\infty} x[k]u[n-k] = x[n] * u[n] = u[n] * x[n].$$

Using the time-convolution property and pair 2 of Table 5.1 yields

$$\sum_{k=0}^n x[k] = u[n] * x[n] \iff \frac{z}{z-1}X[z].$$

No.	$x[n]$	$X[z]$
1	$\delta[n-k]$	z^{-k}
2	$u[n]$	$\frac{z}{z-1}$
3	$nu[n]$	$\frac{z}{(z-1)^2}$
4	$n^2u[n]$	$\frac{z(z+1)}{(z-1)^3}$
5	$n^3u[n]$	$\frac{z(z^2+4z+1)}{(z-1)^4}$
6	$\gamma^n u[n]$	$\frac{z}{z-\gamma}$